MATH I – Week 6

Fall 2026

Derivatives

Sections 3.1-3.4 from textbook

&

Continuity, Differentiability, Left- and Right-sided derivatives, kink points, cusps, vertical tangents¶

Instructor's NOTES (on Moodle)

Lecture presentation prepared and presented by G. Neofotistos

3 DERIVATIVES



Figure 3.1 The Hennessey Venom GT can go from 0 to 200 mph in 14.51 seconds. (credit: modification of work by Codex41, Flickr)

Chapter Outline

3.1 Defining the Derivative
3.2 The Derivative as a Function
3.3 Differentiation Rules
3.4 Derivatives as Rates of Change

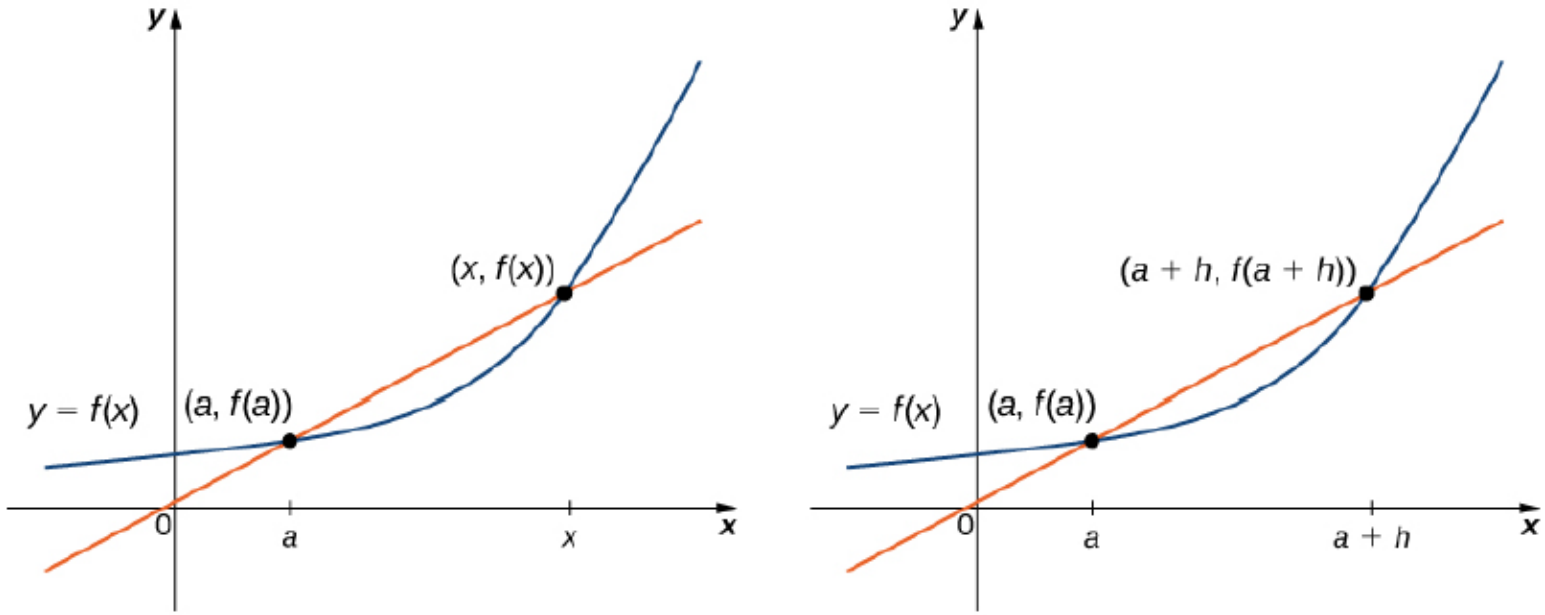
Quick reference: A derivative is a *local sensitivity or rate of change*.

- Time derivatives describe how quickly something evolves;
- spatial derivatives describe how it varies across a system;
- derivatives such as dq/dV describe how one physical quantity responds to another.

No.	Quantity	Derivative definition	Interpretation	SI unit
1	Velocity	$\mathbf{v} = \frac{d\mathbf{r}}{dt}$	Rate of change of position	m/s
2	Speed	$v = \frac{ds}{dt} = \left\lvert \frac{d\mathbf{r}}{dt} \right\rvert$	Rate of change of cumulative distance traveled s	m/s
3	Acceleration	$\mathbf{a} = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2}$	Rate of change of velocity	m/s^2
4	Jerk	$\mathbf{j} = \frac{d\mathbf{a}}{dt}$	Rate of change of acceleration; relevant to vibration and ride comfort	m/s^3



Slope of a secant line



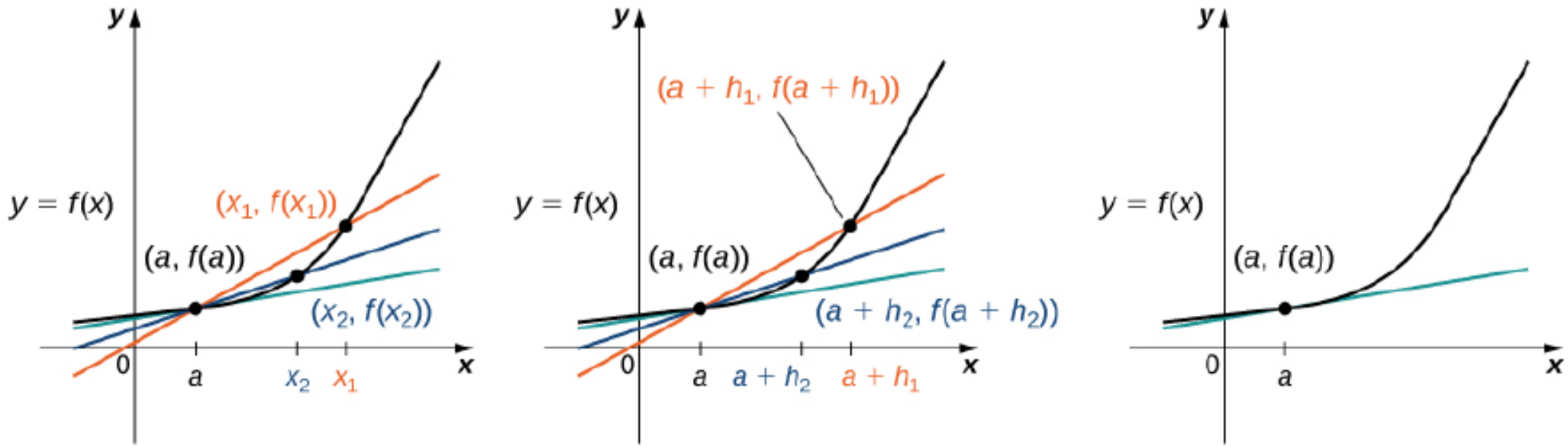
$$m_{sec} = \frac{f(x) - f(a)}{x - a}$$

(a)

$$m_{sec} = \frac{f(a + h) - f(a)}{h}$$

(b)

Figure 3.3 We can calculate the slope of a secant line in either of two ways.



$$m_{\tan} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$$

(a)

$$m_{\tan} = \lim_{h \to 0} \frac{f(a + h) - f(a)}{h}$$

(b)

(c)

Figure 3.4 The secant lines approach the tangent line (shown in green) as the second point approaches the first.

Formally we may define the tangent line to the graph of a function as follows.

Definition

Let $f(x)$ be a function defined in an open interval containing a. The *tangent line* to $f(x)$ at a is the line passing through the point $(a, f(a))$ having slope

$$m_{\tan} = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \tag{3.3}$$

provided this limit exists.

Equivalently, we may define the tangent line to $f(x)$ at a to be the line passing through the point $(a, f(a))$ having slope

$$m_{\tan} = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \tag{3.4}$$

provided this limit exists.

EXAMPLE 3.1

Finding a Tangent Line

Find an equation of the line tangent to the graph of $f(x) = x^2$ at $x = 3$.

Solution

First find the slope of the tangent line. In this example, use Equation 3.3.

$$\begin{aligned} m_{\tan} &= \lim_{x \to 3} \frac{f(x) - f(3)}{x - 3} && \text{Apply the definition.} \\ &= \lim_{x \to 3} \frac{x^2 - 9}{x - 3} && \text{Substitute } f(x) = x^2 \text{ and } f(3) = 9. \\ &= \lim_{x \to 3} \frac{(x-3)(x+3)}{x-3} = \lim_{x \to 3} (x + 3) = 6 && \text{Factor the numerator to evaluate the limit.} \end{aligned}$$

Next, find a point on the tangent line. Since the line is tangent to the graph of $f(x)$ at $x = 3$, it passes through the point $(3, f(3))$. We have $f(3) = 9$, so the tangent line passes through the point $(3, 9)$.

Using the point-slope equation of the line with the slope $m = 6$ and the point $(3, 9)$, we obtain the line $y - 9 = 6(x - 3)$. Simplifying, we have $y = 6x - 9$. The graph of $f(x) = x^2$ and its tangent line at 3 are shown in Figure 3.6.

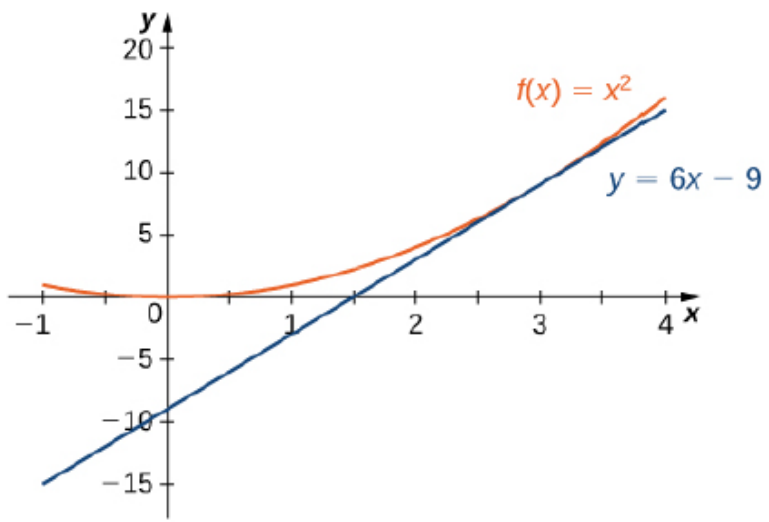


Figure 3.6 The tangent line to $f(x)$ at $x = 3$.

EXAMPLE 3.3

Finding the Equation of a Tangent Line

Find an equation of the line tangent to the graph of $f(x) = 1/x$ at $x = 2$.

Solution

We can use Equation 3.3, but as we have seen, the results are the same if we use Equation 3.4.

$$
\begin{aligned}
m_{\tan} &= \lim_{x \to 2} \frac{f(x) - f(2)}{x - 2} && \text{Apply the definition.} \\
&= \lim_{x \to 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2} && \text{Substitute } f(x) = \tfrac{1}{x} \text{ and } f(2) = \tfrac{1}{2}. \\
&= \lim_{x \to 2} \frac{\frac{1}{x} - \frac{1}{2}}{x - 2} \cdot \frac{2x}{2x} && \text{Multiply numerator and denominator by } 2x \text{ to simplify fractions.} \\
&= \lim_{x \to 2} \frac{(2 - x)}{(x - 2)(2x)} && \text{Simplify.} \\
&= \lim_{x \to 2} \frac{-1}{2x} && \text{Simplify using } \frac{2 - x}{x - 2} = -1, \text{ for } x \neq 2. \\
&= -\frac{1}{4} && \text{Evaluate the limit.}
\end{aligned}
$$

We now know that the slope of the tangent line is $-\frac{1}{4}$. To find an equation of the tangent line, we also need a point on the line. We know that $f(2) = \frac{1}{2}$. Since the tangent line passes through the point $(2, \frac{1}{2})$ we can use the point-slope equation of a line to find an equation of the tangent line. Thus the tangent line has the equation $y = -\frac{1}{4}x + 1$. The graphs of $f(x) = \frac{1}{x}$ and $y = -\frac{1}{4}x + 1$ are shown in Figure 3.7.

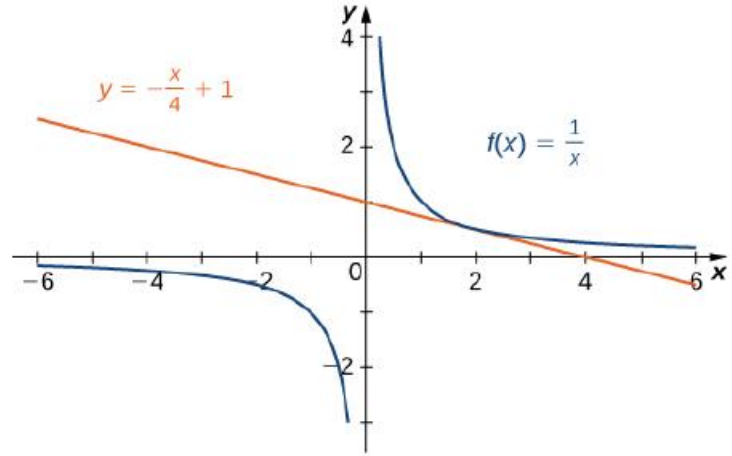


Figure 3.7 The line is tangent to $f(x)$ at $x = 2$.

The Derivative of a Function at a Point

The type of limit we compute in order to find the slope of the line tangent to a function at a point occurs in many applications across many disciplines. These applications include velocity and acceleration in physics, marginal profit functions in business, and growth rates in biology. This limit occurs so frequently that we give this value a special name: the **derivative**. The process of finding a derivative is called **differentiation**.

Definition

Let $f(x)$ be a function defined in an open interval containing a. The derivative of the function $f(x)$ at a, denoted by $f'(a)$, is defined by

$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \tag{3.5}$$

provided this limit exists.

Alternatively, we may also define the derivative of $f(x)$ at a as

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}. \tag{3.6}$$

Finding the derivative of a function at a point

EXAMPLE 3.5

Finding a Derivative

For $f(x) = 3x^2 - 4x + 1$, find $f'(2)$ by using Equation 3.5.

Solution

Substitute the given function and value directly into the equation.

$$\begin{aligned} f'(2) &= \lim_{x \to 2} \frac{f(x) - f(2)}{x - 2} && \text{Apply the definition.} \\ &= \lim_{x \to 2} \frac{\left(3x^2 - 4x + 1\right) - 5}{x - 2} && \text{Substitute } f(x) = 3x^2 - 4x + 1 \text{ and } f(2) = 5. \\ &= \lim_{x \to 2} \frac{(x - 2)(3x + 2)}{x - 2} && \text{Simplify and factor the numerator.} \\ &= \lim_{x \to 2} (3x + 2) && \text{Cancel the common factor.} \\ &= 8 && \text{Evaluate the limit.} \end{aligned}$$

EXAMPLE 3.6

Revisiting the Derivative

For $f(x) = 3x^2 - 4x + 1$, find $f'(2)$ by using Equation 3.6.

Solution

Using this equation, we can substitute two values of the function into the equation, and we should get the same value as in Example 3.5.

$$\begin{aligned} f'(2) &= \lim_{h \to 0} \frac{f(2 + h) - f(2)}{h} && \text{Apply the definition.} \\ &= \lim_{h \to 0} \frac{(3(2 + h)^2 - 4(2 + h) + 1) - 5}{h} && \text{Substitute } f(2) = 5 \text{ and } f(2 + h) = 3(2 + h)^2 - 4(2 + h) + 1. \\ &= \lim_{h \to 0} \frac{3h^2 + 8h}{h} && \text{Simplify the numerator.} \\ &= \lim_{h \to 0} \frac{h(3h + 8)}{h} && \text{Factor the numerator.} \\ &= \lim_{h \to 0} (3h + 8) && \text{Cancel the common factor.} \\ &= 8 && \text{Evaluate the limit.} \end{aligned}$$

https://www.youtube.com/shorts/UOTmbY2ejkg

Velocities and Rates of Change

Now that we can evaluate a derivative, we can use it in velocity applications. Recall that if $s(t)$ is the position of an object moving along a coordinate axis, the average velocity of the object over a time interval $[a, t]$ if $t > a$ or $[t, a]$ if $t < a$ is given by the difference quotient

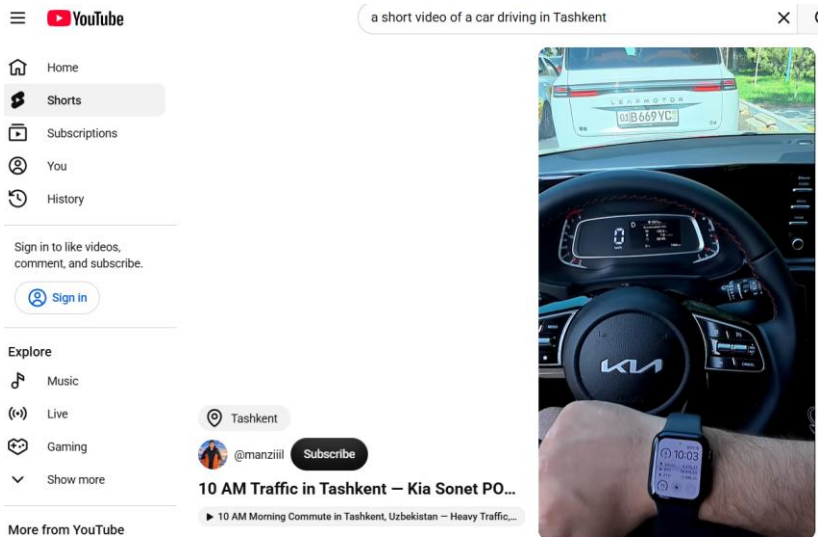
$$v_{\text{ave}} = \frac{s(t) - s(a)}{t - a}. \tag{3.7}$$

As the values of t approach a, the values of v_{ave} approach the value we call the instantaneous velocity at a. That is, instantaneous velocity at a, denoted $v(a)$, is given by

$$v(a) = s'(a) = \lim_{t \to a} \frac{s(t) - s(a)}{t - a}. \tag{3.8}$$

Definition

The **instantaneous rate of change** of a function $f(x)$ at a value a is its derivative $f'(a)$.



EXAMPLE 3.8

Chapter Opener: Estimating Rate of Change of Velocity



Figure 3.10 (credit: modification of work by Codex41, Flickr)

Reaching a top speed of 270.49 mph, the Hennessey Venom GT is one of the fastest cars in the world. In tests it went from 0 to 60 mph in 3.05 seconds, from 0 to 100 mph in 5.88 seconds, from 0 to 200 mph in 14.51 seconds, and from 0 to 229.9 mph in 19.96 seconds. Use this data to draw a conclusion about the rate of change of velocity (that is, its acceleration) as it approaches 229.9 mph. Does the rate at which the car is accelerating appear to be increasing, decreasing, or constant?

Solution

First observe that 60 mph = 88 ft/s, 100 mph ≈ 146.67 ft/s, 200 mph ≈ 293.33 ft/s, and 229.9 mph ≈ 337.19 ft/s. We can summarize the information in a table.

t	$v(t)$
0	0
3.05	88
5.88	146.67
14.51	293.33
19.96	337.19

Table 3.2 $v(t)$ at different values of t

Now compute the average acceleration of the car in feet per second per second on intervals of the form $[t, 19.96]$ as t approaches 19.96, as shown in the following table.

t	$\frac{v(t)-v(19.96)}{t-19.96} = \frac{v(t)-337.19}{t-19.96}$
0.0	16.89
3.05	14.74
5.88	13.53
14.51	8.05

Table 3.3 Average acceleration

The rate at which the car is accelerating is decreasing as its velocity approaches 229.9 mph (337.19 ft/s).

EXAMPLE 3.7

Estimating Velocity

A lead weight on a spring is oscillating up and down. Its position at time t with respect to a fixed horizontal line is given by $s(t) = \sin t$ (Figure 3.9). Use a table of values to estimate $v(0)$. Check the estimate by using Equation 3.5.

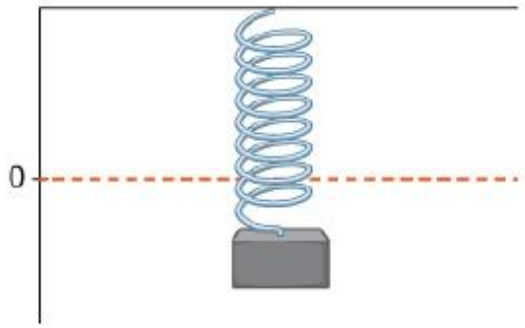


Figure 3.9 A lead weight suspended from a spring in vertical oscillatory motion.

Solution

We can estimate the instantaneous velocity at $t = 0$ by computing a table of average velocities using values of t

By using Equation 3.5, we can see that

$$v(0) = s'(0) = \lim_{t \to 0} \frac{\sin t - \sin 0}{t - 0} = \lim_{t \to 0} \frac{\sin t}{t} = 1.$$

Thus, in fact, $v(0) = 1$.

...by Table and by Equation...

t	$\frac{\sin t - \sin 0}{t - 0} = \frac{\sin t}{t}$
−0.1	0.998334166
−0.01	0.9999833333
−0.001	0.999999833
0.001	0.999999833
0.01	0.9999833333
0.1	0.998334166

Table 3.1 Average velocities using values of t approaching 0

EXAMPLE 3.9

Rate of Change of Temperature

A homeowner sets the thermostat so that the temperature in the house begins to drop from $70°\text{F}$ at 9 p.m., reaches a low of $60°$ during the night, and rises back to $70°$ by 7 a.m. the next morning. Suppose that the temperature in the house is given by $T(t) = 0.4t^2 - 4t + 70$ for $0 \leq t \leq 10$, where t is the number of hours past 9 p.m. Find the instantaneous rate of change of the temperature at midnight.

Solution

Since midnight is 3 hours past 9 p.m., we want to compute $T'(3)$. Refer to Equation 3.5.

$$
\begin{aligned}
T'(3) &= \lim_{t \to 3} \frac{T(t) - T(3)}{t - 3} && \text{Apply the definition.} \\
&= \lim_{t \to 3} \frac{0.4t^2 - 4t + 70 - 61.6}{t - 3} && \text{Substitute } T(t) = 0.4t^2 - 4t + 70 \text{ and } T(3) = 61.6. \\
&= \lim_{t \to 3} \frac{0.4t^2 - 4t + 8.4}{t - 3} && \text{Simplify.} \\
&= \lim_{t \to 3} \frac{0.4(t - 3)(t - 7)}{t - 3} && = \lim_{t \to 3} \frac{0.4(t - 3)(t - 7)}{t - 3} \\
&= \lim_{t \to 3} 0.4(t - 7) && \text{Cancel.} \\
&= -1.6 && \text{Evaluate the limit.}
\end{aligned}
$$

The instantaneous rate of change of the temperature at midnight is $-1.6°\text{F}$ per hour.

3.2 The Derivative as a Function

Learning Objectives

- **3.2.1** Define the derivative function of a given function.
- **3.2.2** Graph a derivative function from the graph of a given function.
- **3.2.3** State the connection between derivatives and continuity.
- **3.2.4** Describe three conditions for when a function does not have a derivative.
- **3.2.5** Explain the meaning of a higher-order derivative.

As we have seen, the derivative of a function at a given point gives us the rate of change or slope of the tangent line to the function at that point. If we differentiate a position function at a given time, we obtain the velocity at that time. It seems reasonable to conclude that knowing the derivative of the function at every point would produce valuable information about the behavior of the function. However, the process of finding the derivative at even a handful of values using the techniques of the preceding section would quickly become quite tedious. In this section we define the derivative function and learn a process for finding it.

Derivative Functions

The derivative function gives the derivative of a function at each point in the domain of the original function for which the derivative is defined. We can formally define a derivative function as follows.

Definition

Let f be a function. The **derivative function**, denoted by f', is the function whose domain consists of those values of x such that the following limit exists:

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}. \tag{3.9}$$

A function $f(x)$ is said to be **differentiable at** a if $f'(a)$ exists. More generally, a function is said to be **differentiable on** S if it is differentiable at every point in an open set S, and a **differentiable function** is one in which $f'(x)$ exists on its domain.

EXAMPLE 3.11

Finding the Derivative of a Square-Root Function

Find the derivative of $f(x) = \sqrt{x}$.

Solution

Start directly with the definition of the derivative function. Use Equation 3.1.

$$f'(x) = \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h}$$

Substitute $f(x+h) = \sqrt{x+h}$ and $f(x) = \sqrt{x}$ into $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$.

$$= \lim_{h \to 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} \cdot \frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}}$$

Multiply numerator and denominator by $\sqrt{x+h} + \sqrt{x}$ without distributing in the denominator.

$$= \lim_{h \to 0} \frac{h}{h(\sqrt{x+h} + \sqrt{x})}$$

Multiply the numerators and simplify.

$$= \lim_{h \to 0} \frac{1}{(\sqrt{x+h} + \sqrt{x})}$$

Cancel the h.

$$= \frac{1}{2\sqrt{x}}$$

Evaluate the limit.

EXAMPLE 3.12

Finding the Derivative of a Quadratic Function

Find the derivative of the function $f(x) = x^2 - 2x$.

Solution

Follow the same procedure here, but without having to multiply by the conjugate.

$$f'(x) = \lim_{h \to 0} \frac{((x+h)^2 - 2(x+h)) - (x^2 - 2x)}{h}$$

Substitute $f(x+h) = (x+h)^2 - 2(x+h)$ and $f(x) = x^2 - 2x$ into $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$.

$$= \lim_{h \to 0} \frac{x^2 + 2xh + h^2 - 2x - 2h - x^2 + 2x}{h}$$

Expand $(x+h)^2 - 2(x+h)$.

$$= \lim_{h \to 0} \frac{2xh - 2h + h^2}{h}$$

Simplify.

$$= \lim_{h \to 0} \frac{h(2x - 2 + h)}{h}$$

Factor out h from the numerator.

$$= \lim_{h \to 0} (2x - 2 + h)$$

Cancel the common factor of h.

$$= 2x - 2$$

Evaluate the limit.

Notations

We use a variety of different notations to express the derivative of a function. In Example 3.12 we showed that if $f(x) = x^2 - 2x$, then $f'(x) = 2x - 2$. If we had expressed this function in the form $y = x^2 - 2x$, we could have expressed the derivative as $y' = 2x - 2$ or $\frac{dy}{dx} = 2x - 2$. We could have conveyed the same information by writing $\frac{d}{dx}\left(x^2 - 2x\right) = 2x - 2$. Thus, for the function $y = f(x)$, each of the following notations represents the derivative of $f(x)$:

$$f'(x), \quad \frac{dy}{dx}, \quad y', \quad \frac{d}{dx}(f(x)).$$

In place of $f'(a)$ we may also use $\left.\frac{dy}{dx}\right|_{x=a}$ Use of the $\frac{dy}{dx}$ notation (called Leibniz notation) is quite common in engineering and physics. To understand this notation better, recall that the derivative of a function at a point is the limit of the slopes of secant lines as the secant lines approach the tangent line. The slopes of these secant lines are often expressed in the form $\frac{\Delta y}{\Delta x}$ where Δy is the difference in the y values corresponding to the difference in the x values, which are expressed as Δx (Figure 3.11). Thus the derivative, which can be thought of as the instantaneous rate of change of y with respect to x, is expressed as

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}.$$

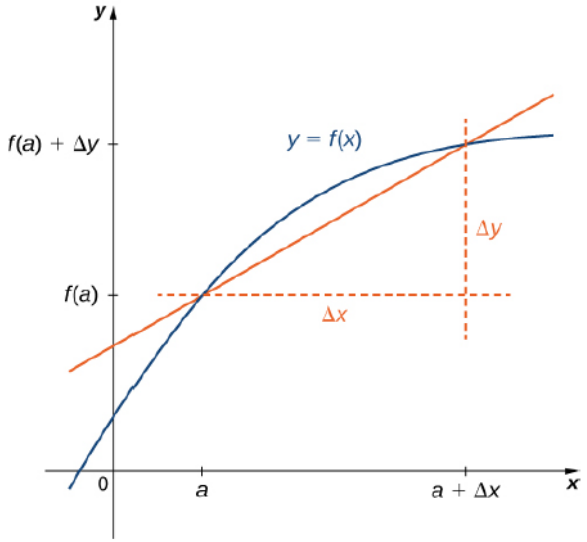


Figure 3.11 The derivative is expressed as $\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}$.

Graphing a derivative
2 examples…

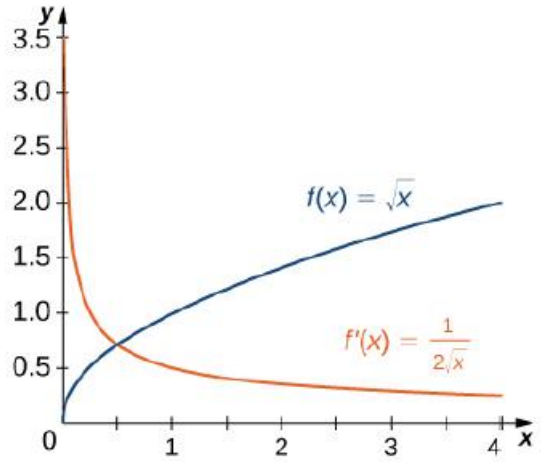


Figure 3.12 The derivative $f'(x)$ is positive everywhere because the function $f(x)$ is increasing.

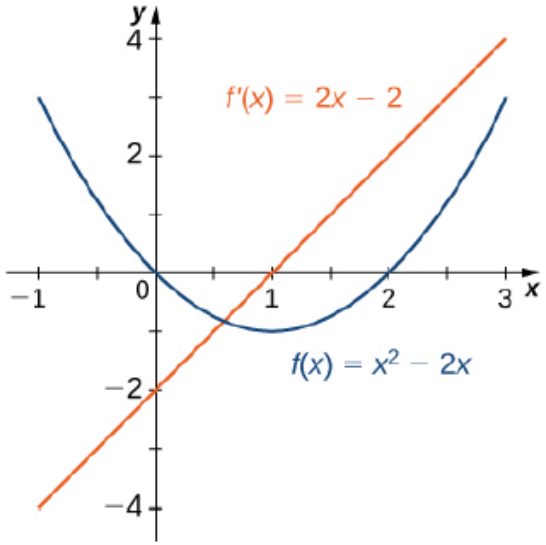
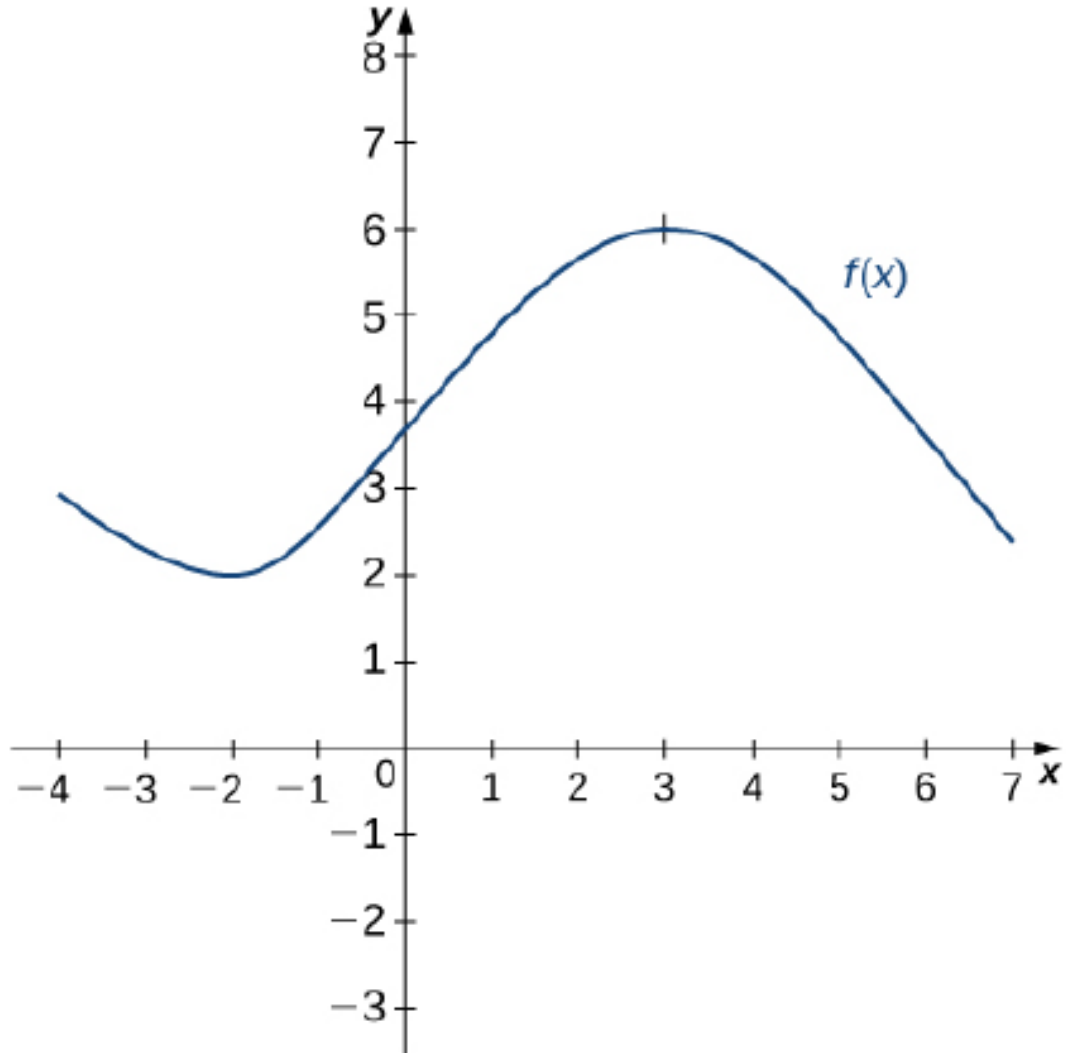


Figure 3.13 The derivative $f'(x) < 0$ where the function $f(x)$ is decreasing and $f'(x) > 0$ where $f(x)$ is increasing. The derivative is zero where the function has a horizontal tangent.

EXAMPLE 3.13

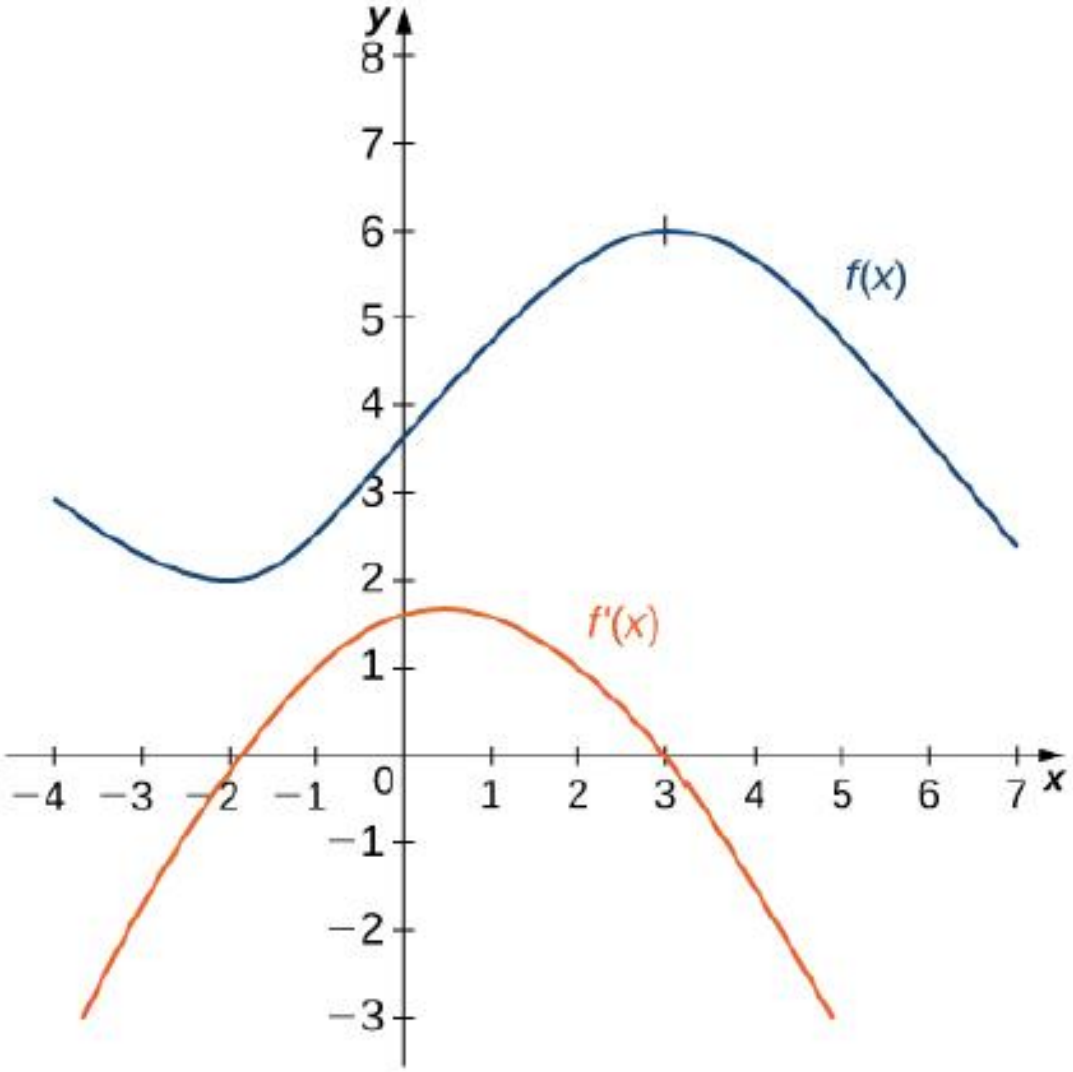
Sketching a Derivative Using a Function

Use the following graph of $f(x)$ to sketch a graph of $f'(x)$.



In-Class activity

Solution



Derivatives and Continuity

Now that we can graph a derivative, let's examine the behavior of the graphs. First, we consider the relationship between differentiability and continuity. We will see that if a function is differentiable at a point, it must be continuous there; however, a function that is continuous at a point need not be differentiable at that point. In fact, a function may be continuous at a point and fail to be differentiable at the point for one of several reasons.

Theorem 3.1

Differentiability Implies Continuity
Let $f(x)$ be a function and a be in its domain. If $f(x)$ is differentiable at a, then f is continuous at a.

Proof

If $f(x)$ is differentiable at a, then $f'(a)$ exists and

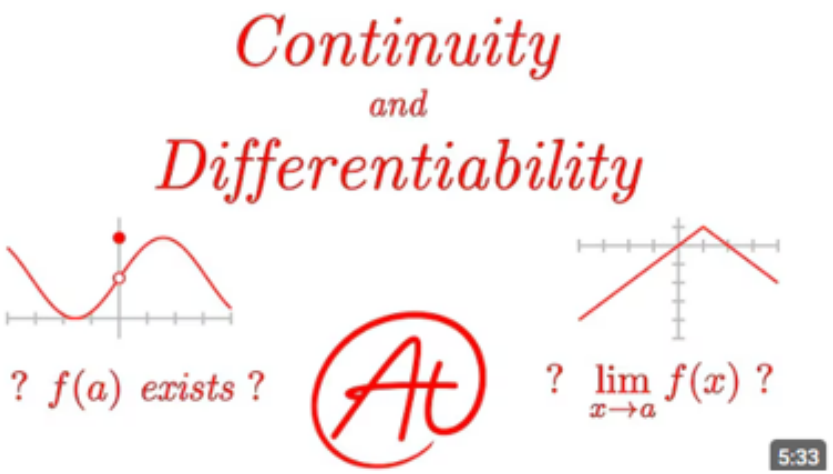
$$f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

We want to show that $f(x)$ is continuous at a by showing that $\lim_{x \to a} f(x) = f(a)$. Thus,

$$\begin{aligned}
\lim_{x \to a} f(x) &= \lim_{x \to a} (f(x) - f(a) + f(a)) \\
&= \lim_{x \to a} \left(\frac{f(x) - f(a)}{x - a} \cdot (x - a) + f(a) \right) && \text{Multiply and divide } f(x) - f(a) \text{ by } x - a. \\
&= \left(\lim_{x \to a} \frac{f(x) - f(a)}{x - a} \right) \cdot \left(\lim_{x \to a} (x - a) \right) + \lim_{x \to a} f(a) \\
&= f'(a) \cdot 0 + f(a) \\
&= f(a).
\end{aligned}$$

Therefore, since $f(a)$ is defined and $\lim_{x \to a} f(x) = f(a)$, we conclude that f is continuous at a.

https://www.youtube.com/watch?v=nRv1lPqrx3k



Continuity and Differentiability EXPLAINED with Examples

288K • 4y ago

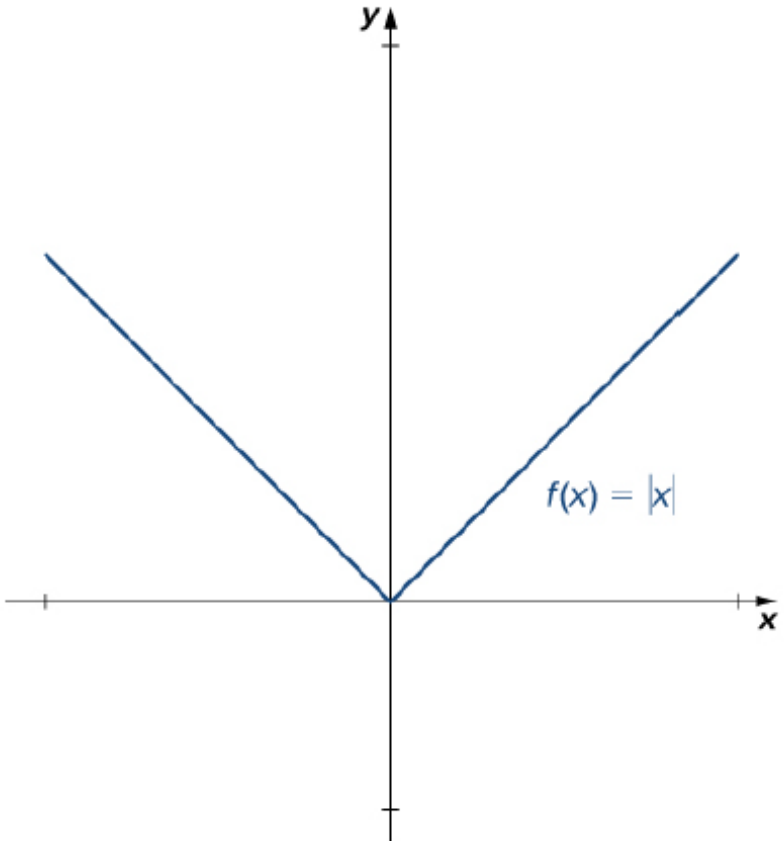
Ace Tutors

Learn the meaning behind **Continuity** and Differentiability. This video explains what **Continuity** is and the 3 rules for a **function**

4K College

In-Class activity

Is the function |x|
Continuous? Differentiable?
at x=0



continuous everywhere; however, $f'(0)$ is undefined. This observation leads us to believe that continuity does not imply differentiability. Let's explore further. For $f(x) = |x|$,

$$f'(0) = \lim_{x \to 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \to 0} \frac{|x| - |0|}{x - 0} = \lim_{x \to 0} \frac{|x|}{x}.$$

This limit does not exist because

$$\lim_{x \to 0^-} \frac{|x|}{x} = -1 \text{ and } \lim_{x \to 0^+} \frac{|x|}{x} = 1.$$

See Figure 3.14.

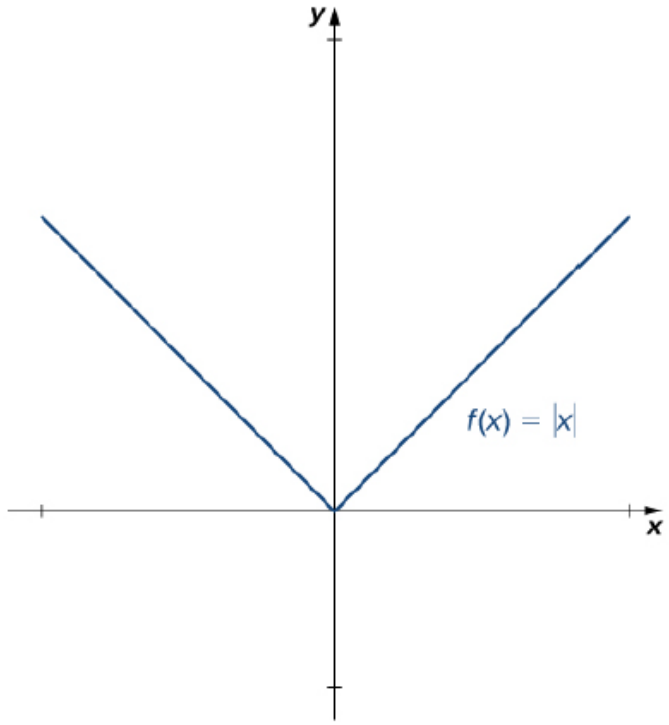


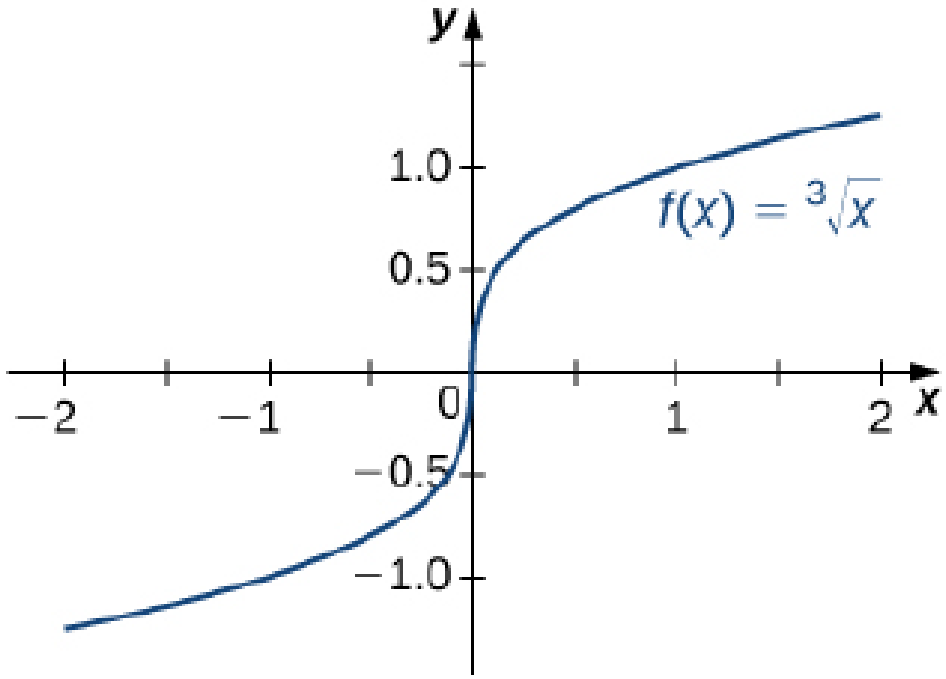
Figure 3.14 The function $f(x) = |x|$ is continuous at 0 but is not differentiable at 0.

Solution

$$f(x) = \sqrt[3]{x}$$

In-Class activity

Is this function
Continuous? Differentiable?
at x=0



Solution

Let's consider some additional situations in which a continuous function fails to be differentiable. Consider the function $f(x) = \sqrt[3]{x}$:

$$f'(0) = \lim_{x \to 0} \frac{\sqrt[3]{x} - 0}{x - 0} = \lim_{x \to 0} \frac{1}{\sqrt[3]{x^2}} = +\infty.$$

Thus $f'(0)$ does not exist. A quick look at the graph of $f(x) = \sqrt[3]{x}$ clarifies the situation. The function has a vertical tangent line at 0 (Figure 3.15).

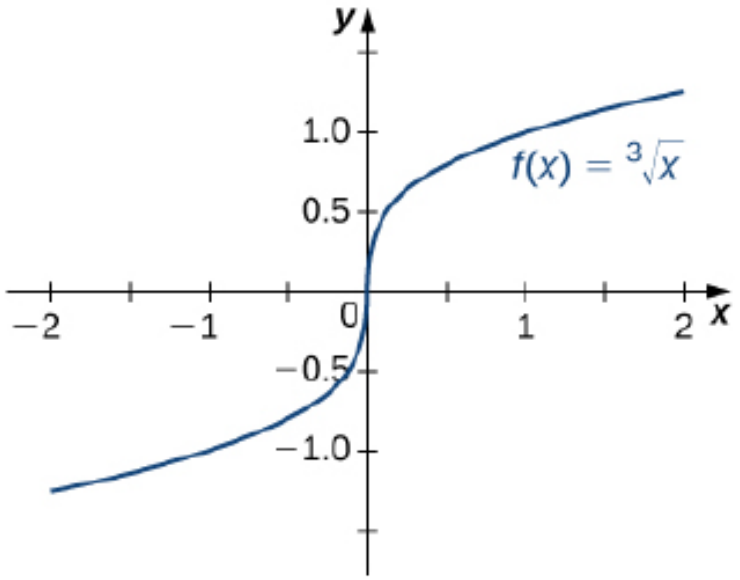


Figure 3.15 The function $f(x) = \sqrt[3]{x}$ has a vertical tangent at $x = 0$. It is continuous at 0 but is not differentiable at 0.

2. Left- and right-hand derivatives

From the NOTES...

We can approach a point from either side:

$$\boxed{f'_-(a) = \lim_{h \to 0^-} \frac{f(a+h) - f(a)}{h}} \qquad \text{(left-hand derivative)}$$

$$\boxed{f'_+(a) = \lim_{h \to 0^+} \frac{f(a+h) - f(a)}{h}} \qquad \text{(right-hand derivative).}$$

At an interior point of the domain,

$$\boxed{f'(a) \text{ exists} \iff f'_-(a) = f'_+(a) \text{ and both are finite.}}$$

For example, if $f(x) = x^2$, then at $a = 0$,

$\frac{f(h)-f(0)}{h} = \frac{h^2}{h} = h.$

From either side, $h \to 0$, so $f'(0) = 0$: the tangent is horizontal.

Slopes approaching the origin

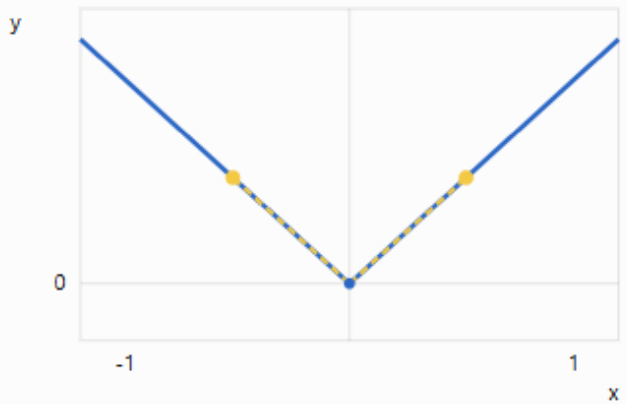
Distance from the origin, h: 0.52

Smooth: $y = x^2$



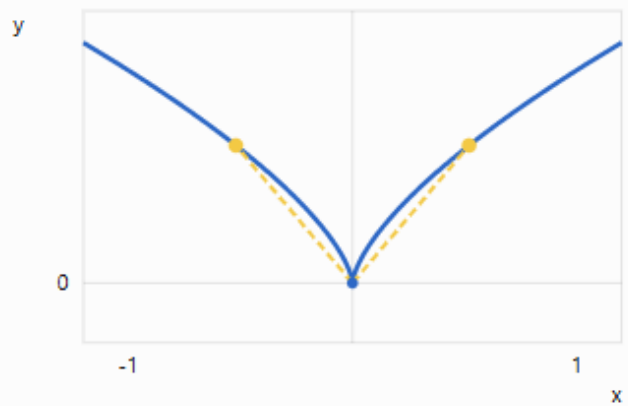
Left slope: -0.52 · Right slope: 0.52

Kink: $y = |x|$



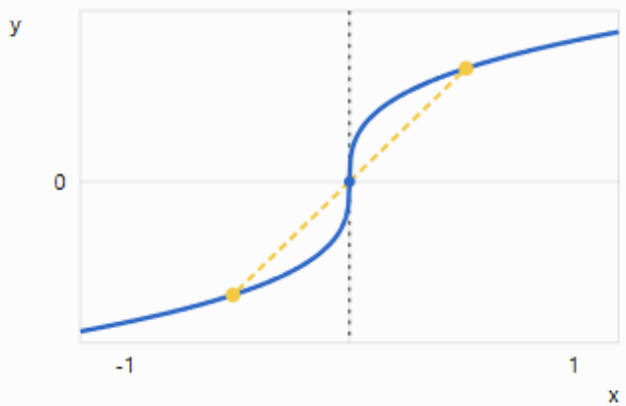
Left slope: -1.00 · Right slope: 1.00

Cusp: $y = |x|^{2/3}$



Left slope: -1.24 · Right slope: 1.24

Vertical tangent: $y = \sqrt[3]{x}$



Left slope: 1.55 · Right slope: 1.55

Solid: curve · Dashed: left and right secants · Dotted: limiting tangent, when it exists

Slopes approaching the origin

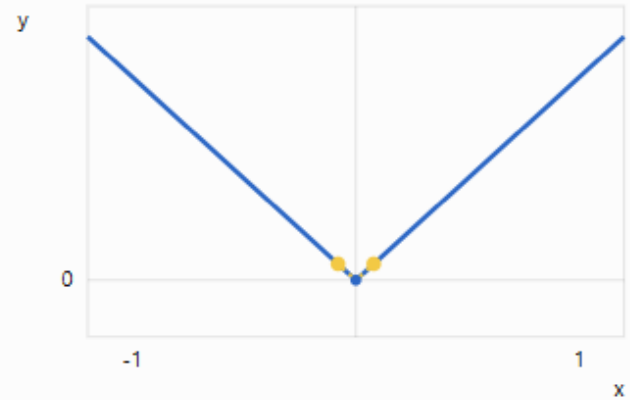
Distance from the origin, h: 0.08

Smooth: $y = x^2$



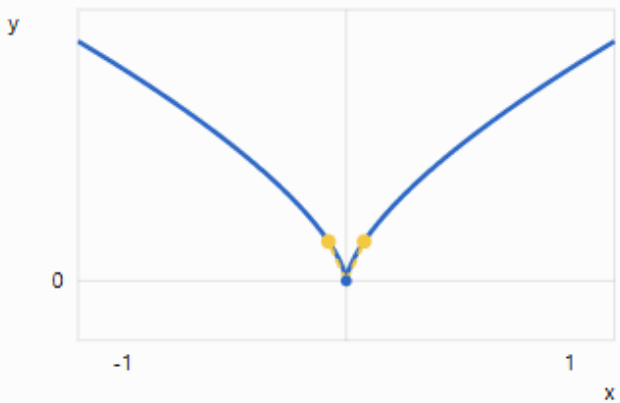
Left slope: -0.08 · Right slope: 0.08

Kink: $y = |x|$



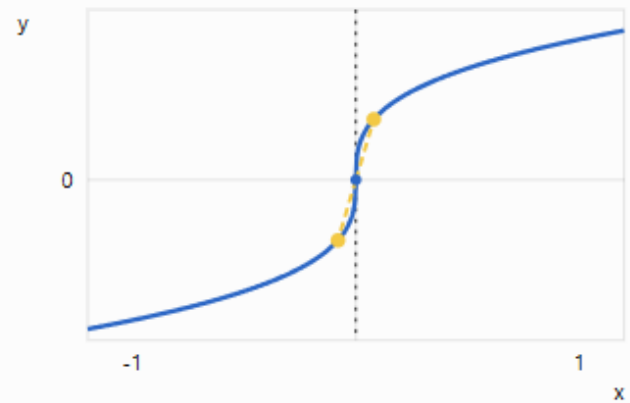
Left slope: -1.00 · Right slope: 1.00

Cusp: $y = |x|^{2/3}$



Left slope: -2.32 · Right slope: 2.32

Vertical tangent: $y = \sqrt[3]{x}$



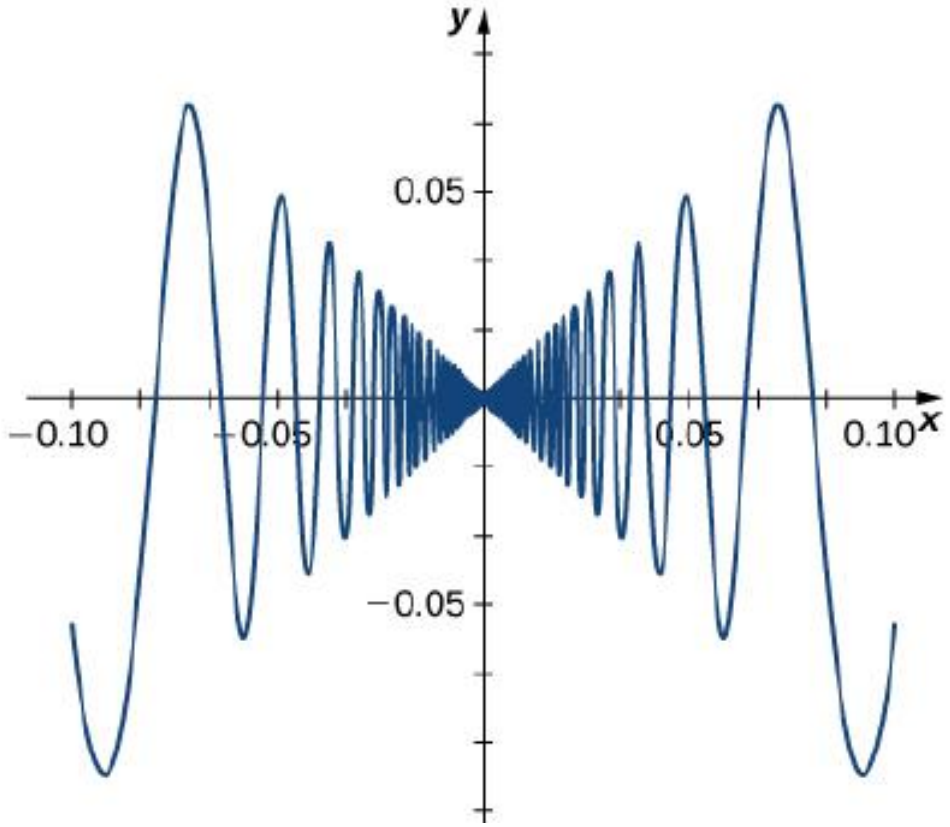
Left slope: 5.39 · Right slope: 5.39

Solid: curve · Dashed: left and right secants · Dotted: limiting tangent, when it exists

$$f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 \text{ if } x = 0 \end{cases}$$

In-Class activity

Is this function
Continuous? Differentiable?
at x=0



Solution

$$f'(0) = \lim_{x \to 0} \frac{x \sin(1/x) - 0}{x - 0} = \lim_{x \to 0} \sin\left(\frac{1}{x}\right).$$

This limit does not exist, essentially because the slopes of the secant lines continuously change direction as they approach zero (Figure 3.16).

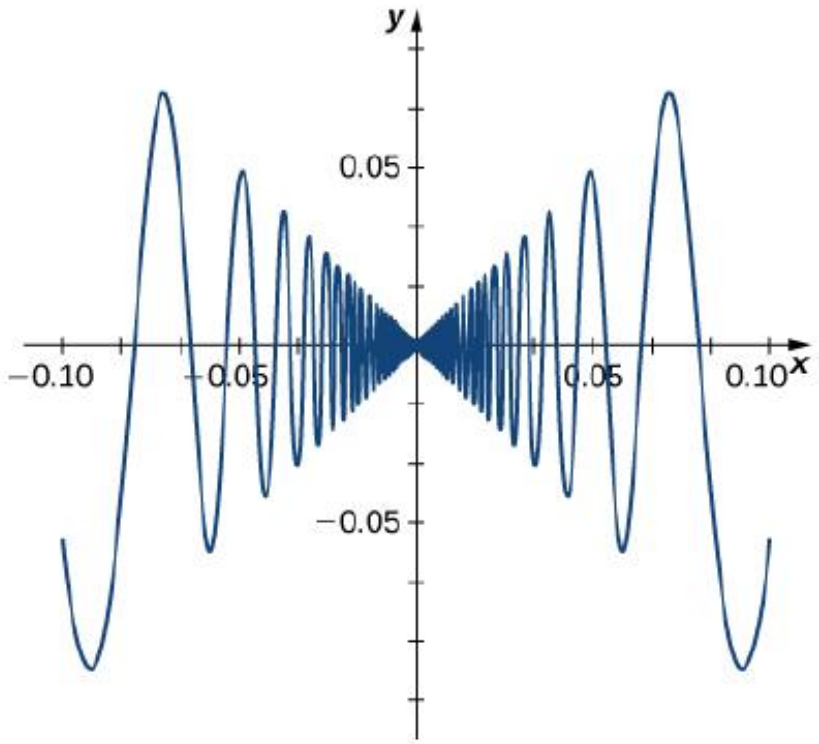


Figure 3.16 The function $f(x) = \begin{cases} x \sin\left(\frac{1}{x}\right) \text{ if } x \neq 0 \\ 0 \text{ if } x = 0 \end{cases}$ is not differentiable at 0.

Higher-Order Derivatives

The derivative of a function is itself a function, so we can find the derivative of a derivative. For example, the derivative of a position function is the rate of change of position, or velocity. The derivative of velocity is the rate of change of velocity, which is acceleration. The new function obtained by differentiating the derivative is called the second derivative. Furthermore, we can continue to take derivatives to obtain the third derivative, fourth derivative, and so on. Collectively, these are referred to as **higher-order derivatives**. The notation for the higher-order derivatives of $y = f(x)$ can be expressed in any of the following forms:

$$f''(x),\ f'''(x), f^{(4)}(x), \ldots, f^{(n)}(x)$$

$$y''(x), y'''(x), y^{(4)}(x), \ldots, y^{(n)}(x)$$

$$\frac{d^2y}{dx^2}, \frac{d^3y}{dx^3}, \frac{d^4y}{dx^4}, \ldots, \frac{d^ny}{dx^n}.$$

It is interesting to note that the notation for $\frac{d^2y}{dx^2}$ may be viewed as an attempt to express $\frac{d}{dx}\left(\frac{dy}{dx}\right)$ more compactly. Analogously, $\frac{d}{dx}\left(\frac{d}{dx}\left(\frac{dy}{dx}\right)\right) = \frac{d}{dx}\left(\frac{d^2y}{dx^2}\right) = \frac{d^3y}{dx^3}$.

EXAMPLE 3.15

Finding a Second Derivative

For $f(x) = 2x^2 - 3x + 1$, find $f''(x)$.

Solution

First find $f'(x)$.

$$f'(x) = \lim_{h \to 0} \frac{\left(2(x+h)^2 - 3(x+h) + 1\right) - (2x^2 - 3x + 1)}{h}$$

Substitute $f(x) = 2x^2 - 3x + 1$ and $f(x+h) = 2(x+h)^2 - 3(x+h) + 1$ into $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$.

$$= \lim_{h \to 0} \frac{4xh + 2h^2 - 3h}{h}$$

Simplify the numerator.

$$= \lim_{h \to 0} (4x + 2h - 3)$$

Factor out the h in the numerator and cancel with the h in the denominator.

$$= 4x - 3$$

Take the limit.

Next, find $f''(x)$ by taking the derivative of $f'(x) = 4x - 3$.

$$f''(x) = \lim_{h \to 0} \frac{f'(x+h) - f'(x)}{h}$$

Use $f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$ with $f'(x)$ in place of $f(x)$.

$$= \lim_{h \to 0} \frac{(4(x+h) - 3) - (4x - 3)}{h}$$

Substitute $f'(x+h) = 4(x+h) - 3$ and $f'(x) = 4x - 3$.

$$= \lim_{h \to 0} 4$$

Simplify.

$$= 4$$

Take the limit.

3.3 Differentiation Rules

Learning Objectives

- **3.3.1** State the constant, constant multiple, and power rules.
- **3.3.2** Apply the sum and difference rules to combine derivatives.
- **3.3.3** Use the product rule for finding the derivative of a product of functions.
- **3.3.4** Use the quotient rule for finding the derivative of a quotient of functions.
- **3.3.5** Extend the power rule to functions with negative exponents.
- **3.3.6** Combine the differentiation rules to find the derivative of a polynomial or rational function.

Theorem 3.2

The Constant Rule

Let c be a constant.

If $f(x) = c$, then $f'(x) = 0$.

Alternatively, we may express this rule as

$$\frac{d}{dx}(c) = 0.$$

The Constant Rule

We first apply the limit definition of the derivative to find the derivative of the constant function, $f(x) = c$. For this function, both $f(x) = c$ and $f(x + h) = c$, so we obtain the following result:

$$\begin{aligned} f'(x) &= \lim_{h \to 0} \frac{f(x+h) - f(x)}{h} \\ &= \lim_{h \to 0} \frac{c - c}{h} \\ &= \lim_{h \to 0} \frac{0}{h} \\ &= \lim_{h \to 0} 0 = 0. \end{aligned}$$

Theorem 3.3

The Power Rule

Let n be a positive integer. If $f(x) = x^n$, then

$$f'(x) = nx^{n-1}.$$

Alternatively, we may express this rule as

$$\frac{d}{dx}x^n = nx^{n-1}.$$

Theorem 3.4

Sum, Difference, and Constant Multiple Rules

Let $f(x)$ and $g(x)$ be differentiable functions and k be a constant. Then each of the following equations holds.

Sum Rule. The derivative of the sum of a function f and a function g is the same as the sum of the derivative of f and the derivative of g.

$$\frac{d}{dx}(f(x) + g(x)) = \frac{d}{dx}(f(x)) + \frac{d}{dx}(g(x));$$

that is,

$$\text{for } j(x) = f(x) + g(x), j'(x) = f'(x) + g'(x).$$

Difference Rule. The derivative of the difference of a function f and a function g is the same as the difference of the derivative of f and the derivative of g:

$$\frac{d}{dx}(f(x) - g(x)) = \frac{d}{dx}(f(x)) - \frac{d}{dx}(g(x));$$

that is,

$$\text{for } j(x) = f(x) - g(x), j'(x) = f'(x) - g'(x).$$

Constant Multiple Rule. The derivative of a constant k multiplied by a function f is the same as the constant multiplied by the derivative:

$$\frac{d}{dx}(kf(x)) = k\frac{d}{dx}(f(x));$$

that is,

$$\text{for } j(x) = kf(x), j'(x) = kf'(x).$$

In-Class activity

EXAMPLE 3.21

Applying Basic Derivative Rules

Find the derivative of $f(x) = 2x^5 + 7$.

EXAMPLE 3.21

Solution

Applying Basic Derivative Rules

Find the derivative of $f(x) = 2x^5 + 7$.

Solution

We begin by applying the rule for differentiating the sum of two functions, followed by the rules for differentiating constant multiples of functions and the rule for differentiating powers. To better understand the sequence in which the differentiation rules are applied, we use Leibniz notation throughout the solution:

$$
\begin{aligned}
f'(x) &= \frac{d}{dx}\left(2x^5 + 7\right) \\
&= \frac{d}{dx}\left(2x^5\right) + \frac{d}{dx}(7) && \text{Apply the sum rule.} \\
&= 2\frac{d}{dx}\left(x^5\right) + \frac{d}{dx}(7) && \text{Apply the constant multiple rule.} \\
&= 2\left(5x^4\right) + 0 && \text{Apply the power rule and the constant rule.} \\
&= 10x^4. && \text{Simplify.}
\end{aligned}
$$

Theorem 3.5

Product Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}(f(x)\,g(x)) = \frac{d}{dx}(f(x)) \cdot g(x) + \frac{d}{dx}(g(x)) \cdot f(x).$$

Theorem 3.6

The Quotient Rule

Let $f(x)$ and $g(x)$ be differentiable functions. Then

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{\frac{d}{dx}(f(x)) \cdot g(x) - \frac{d}{dx}(g(x)) \cdot f(x)}{(g(x))^2}.$$

That is,

$$\text{if } j(x) = \frac{f(x)}{g(x)}, \text{ then } j'(x) = \frac{f'(x)\,g(x) - g'(x)\,f(x)}{(g(x))^2}.$$

In-Class activity

Theorem 3.7

Extended Power Rule

If k is a negative integer, then

$$\frac{d}{dx}\left(x^k\right) = kx^{k-1}.$$

Using the Extended Power Rule

Find $\frac{d}{dx}\left(x^{-4}\right)$.

Solution

EXAMPLE 3.26

Using the Extended Power Rule

Find $\frac{d}{dx}\left(x^{-4}\right)$.

Solution

By applying the extended power rule with $k = -4$, we obtain

$$\frac{d}{dx}\left(x^{-4}\right) = -4x^{-4-1} = -4x^{-5}.$$

In-Class activity

Using the Extended Power Rule and the Constant Multiple Rule

Use the extended power rule and the constant multiple rule to find the derivative of $f(x) = \frac{6}{x^2}$.

Solution

EXAMPLE 3.27

Using the Extended Power Rule and the Constant Multiple Rule

Use the extended power rule and the constant multiple rule to find the derivative of $f(x) = \frac{6}{x^2}$.

Solution

It may seem tempting to use the quotient rule to find this derivative, and it would certainly not be incorrect to do so. However, it is far easier to differentiate this function by first rewriting it as $f(x) = 6x^{-2}$.

$$
\begin{aligned}
f'(x) &= \frac{d}{dx}\left(\frac{6}{x^2}\right) = \frac{d}{dx}\left(6x^{-2}\right) && \text{Rewrite } \frac{6}{x^2} \text{ as } 6x^{-2}. \\
&= 6\frac{d}{dx}(x^{-2}) && \text{Apply the constant multiple rule.} \\
&= 6(-2x^{-3}) && \text{Use the extended power rule to differentiate } x^{-2}. \\
&= -12x^{-3} && \text{Simplify.}
\end{aligned}
$$

EXAMPLE 3.28

Combining Differentiation Rules

For $k(x) = 3h(x) + x^2 g(x)$, find $k'(x)$.

Solution

Finding this derivative requires the sum rule, the constant multiple rule, and the product rule.

$$k'(x) = \frac{d}{dx}\left(3h(x) + x^2 g(x)\right) = \frac{d}{dx}(3h(x)) + \frac{d}{dx}\left(x^2 g(x)\right)$$

Apply the sum rule.

$$= 3\frac{d}{dx}(h(x)) + \left(\frac{d}{dx}\left(x^2\right) g(x) + \frac{d}{dx}(g(x))\, x^2\right)$$

Apply the constant multiple rule to differentiate $3h(x)$ and the product rule to differentiate $x^2 g(x)$.

$$= 3h'(x) + 2xg(x) + x^2 g'(x)$$

$$f(x) = x^3 - 7x^2 + 8x + 1$$

In-Class activity

a) Find the derivative of f(x)

b) Find the zeros of the derivative

EXAMPLE 3.31

Determining Where a Function Has a Horizontal Tangent

Determine the values of x for which $f(x) = x^3 - 7x^2 + 8x + 1$ has a horizontal tangent line.

Solution

Solution

To find the values of x for which $f(x)$ has a horizontal tangent line, we must solve $f'(x) = 0$. Since

$$f'(x) = 3x^2 - 14x + 8 = (3x - 2)(x - 4),$$

we must solve $(3x - 2)(x - 4) = 0$. Thus we see that the function has horizontal tangent lines at $x = \frac{2}{3}$ and $x = 4$ as shown in the following graph.

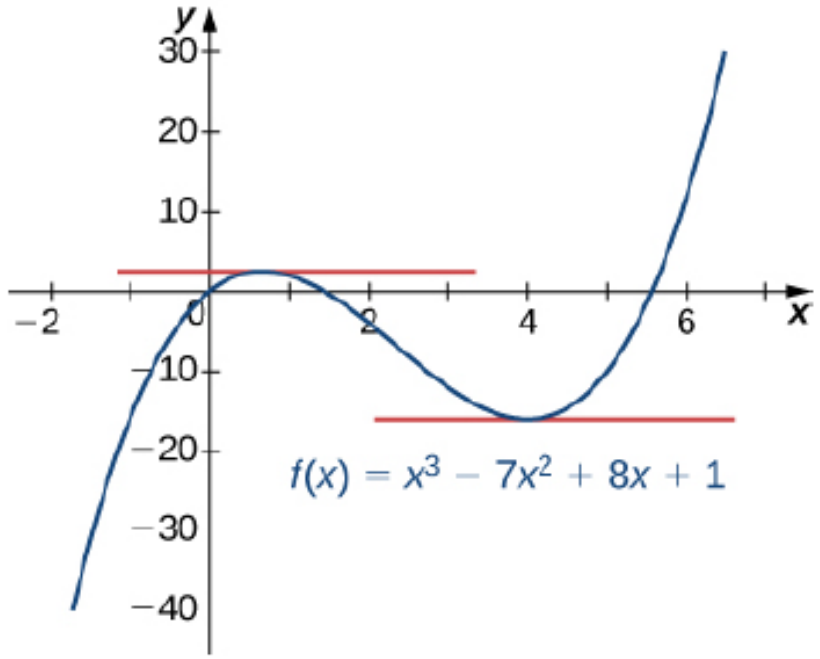


Figure 3.19 This function has horizontal tangent lines at $x = 2/3$ and $x = 4$.

Amount of Change Formula

One application for derivatives is to estimate an unknown value of a function at a point by using a known value of a function at some given point together with its rate of change at the given point. If $f(x)$ is a function defined on an interval $[a, a+h]$, then the **amount of change** of $f(x)$ over the interval is the change in the y values of the function over that interval and is given by

$$f(a+h) - f(a).$$

The **average rate of change** of the function f over that same interval is the ratio of the amount of change over that interval to the corresponding change in the x values. It is given by

$$\frac{f(a+h) - f(a)}{h}.$$

As we already know, the instantaneous rate of change of $f(x)$ at a is its derivative

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

For small enough values of h, $f'(a) \approx \frac{f(a+h)-f(a)}{h}$. We can then solve for $f(a+h)$ to get the amount of change formula:

$$f(a+h) \approx f(a) + f'(a)h. \tag{3.10}$$

EXAMPLE 3.33

Estimating the Value of a Function

If $f(3) = 2$ and $f'(3) = 5$, estimate $f(3.2)$.

Solution

Begin by finding h. We have $h = 3.2 - 3 = 0.2$. Thus,

$$f(3.2) = f(3+0.2) \approx f(3) + (0.2) f'(3) = 2 + 0.2(5) = 3.$$

Motion along a Line

Another use for the derivative is to analyze motion along a line. We have described velocity as the rate of change of position. If we take the derivative of the velocity, we can find the acceleration, or the rate of change of velocity. It is also important to introduce the idea of **speed**, which is the magnitude of velocity. Thus, we can state the following mathematical definitions.

Definition

Let $s(t)$ be a function giving the position of an object at time t.

The velocity of the object at time t is given by $v(t) = s'(t)$.

The speed of the object at time t is given by $|v(t)|$.

The acceleration of the object at t is given by $a(t) = v'(t) = s''(t)$.



In-Class activity Exercises

Instructions

- Show your reasoning and intermediate calculations.
- When the derivative definition is requested, do not use differentiation rules.
- Give units in engineering applications.
- Unless stated otherwise, "differentiable" means having a **finite derivative**.
- No chain rule, implicit differentiation, or trigonometric differentiation is required.

1. Derivative at a point

Let $f(x) = x^2 + 3x$.

Using

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h},$$

find $f'(2)$. Explain the geometric meaning of your answer.

Instructions

- Show your reasoning and intermediate calculations.
- When the derivative definition is requested, do not use differentiation rules.
- Give units in engineering applications.
- Unless stated otherwise, "differentiable" means having a **finite derivative**.
- No chain rule, implicit differentiation, or trigonometric differentiation is required.

In-Class activity Exercise 1 Solution

1. Derivative at a point

Let $f(x) = x^2 + 3x$.

Using

$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$,

find $f'(2)$. Explain the geometric meaning of your answer.

1. Derivative at a point

For $f(x) = x^2 + 3x$,

$$f(2) = 10, \qquad f(2+h) = 10 + 7h + h^2.$$

Therefore,

$$f'(2) = \lim_{h \to 0} \frac{7h + h^2}{h} = \lim_{h \to 0} (7 + h) = \boxed{7}.$$

Geometrically, the tangent line at $(2, 10)$ has slope 7.

Exercise 2

2. Derivative as a function

For $f(x) = 3x^2 - 2x + 4$,

use the limit definition to:

a. Find $f'(x)$.
b. Find the slope of the tangent at $x = 1$.
c. Find the equation of that tangent.

2. Derivative as a function

For $f(x) = 3x^2 - 2x + 4$,

use the limit definition to:

a. Find $f'(x)$.
b. Find the slope of the tangent at $x = 1$.
c. Find the equation of that tangent.

Exercise 2 Solution

2. Derivative as a function

For $f(x) = 3x^2 - 2x + 4$,

$$f'(x) = \lim_{h \to 0} \frac{3(x+h)^2 - 2(x+h) + 4 - (3x^2 - 2x + 4)}{h}$$
$$= \lim_{h \to 0} (6x + 3h - 2) = \boxed{6x - 2}.$$

At $x = 1$,

$$f'(1) = 4, \qquad f(1) = 5.$$

The tangent line is

$$y - 5 = 4(x - 1) \quad \Rightarrow \quad \boxed{y = 4x + 1}.$$

Exercise 3

Differentiate:

a. $f(x) = 5x^4 - 3x^3 + 2x - 7.$

b. $g(x) = \frac{x^5}{5} - \frac{3x^2}{2} + 8.$

c. $p(x) = 7 - 4x + x^6.$

Exercise 3 Solution

Differentiate:

a. $f(x) = 5x^4 - 3x^3 + 2x - 7.$

b. $g(x) = \frac{x^5}{5} - \frac{3x^2}{2} + 8.$

c. $p(x) = 7 - 4x + x^6.$

a. $\boxed{f'(x) = 20x^3 - 9x^2 + 2}.$

b. $\boxed{g'(x) = x^4 - 3x}.$

c. $\boxed{p'(x) = 6x^5 - 4}.$

Exercise 4

Rewrite using powers, then differentiate. State the domain of each derivative.

a. $f(x) = \frac{3}{x^2}$.

b. $g(x) = 2\sqrt{x} - \frac{1}{x}$.

c. $p(x) = \sqrt[3]{x} + x^2$.

Exercise 4 Solution

Rewrite using powers, then differentiate. State the domain of each derivative.

a. $f(x) = \frac{3}{x^2}$.

b. $g(x) = 2\sqrt{x} - \frac{1}{x}$.

c. $p(x) = \sqrt[3]{x} + x^2$.

a. $f(x) = 3x^{-2} \quad \Rightarrow \quad \boxed{f'(x) = -6x^{-3} = -\dfrac{6}{x^3}}, \qquad x \neq 0.$

b. $g(x) = 2x^{1/2} - x^{-1} \quad \Rightarrow \quad \boxed{g'(x) = \dfrac{1}{\sqrt{x}} + \dfrac{1}{x^2}}, \qquad x > 0.$

c. $p(x) = x^{1/3} + x^2 \quad \Rightarrow \quad \boxed{p'(x) = \dfrac{1}{3x^{2/3}} + 2x}, \qquad x \neq 0.$

Although p is defined at 0, it has no finite derivative there.

Exercise 5

Let $f(x) = \frac{x^2+1}{x+2}$, $\quad x \neq -2.$

a. Find $f'(x)$ using the quotient rule.

b. Find $f'(0)$.

c. Find the tangent line at $x = 0$.

d. Explain why the function has no derivative at $x = -2$.

Exercise 5 Solution

Let $f(x) = \frac{x^2+1}{x+2}, \qquad x \neq -2.$

a. Find $f'(x)$ using the quotient rule.

b. Find $f'(0)$.

c. Find the tangent line at $x = 0$.

d. Explain why the function has no derivative at $x = -2$.

$$f'(x) = \frac{2x(x+2) - (x^2+1)}{(x+2)^2}$$

$$= \boxed{\frac{x^2+4x-1}{(x+2)^2}}.$$

At 0,

$$f'(0) = -\tfrac{1}{4}, \qquad f(0) = \tfrac{1}{2}.$$

The tangent is

$$\boxed{y = \frac{1}{2} - \frac{x}{4}}.$$

At $x = -2$, the function is undefined, so it has no derivative there.

Exercise 6

For

$$f(x) = 2x^4 - 3x^3 + x^2 - 5x + 6,$$

find

$$f'(x), \quad f''(x), \quad f'''(x), \quad f^{(4)}(x), \quad f^{(5)}(x).$$

Explain why the fifth derivative has its particular value.

Exercise 6 Solution

For

$$f(x) = 2x^4 - 3x^3 + x^2 - 5x + 6,$$

find

$$f'(x), \quad f''(x), \quad f'''(x), \quad f^{(4)}(x), \quad f^{(5)}(x).$$

Explain why the fifth derivative has its particular value.

$$\begin{aligned} f'(x) &= 8x^3 - 9x^2 + 2x - 5, \\ f''(x) &= 24x^2 - 18x + 2, \\ f'''(x) &= 48x - 18, \\ f^{(4)}(x) &= 48, \\ f^{(5)}(x) &= 0. \end{aligned}$$

Each differentiation reduces the degree by one until a constant is reached. Its derivative is zero.

Exercise 7

Differentiate

$$f(x) = \sqrt{x}(x^2 + 4), \qquad x > 0.$$

a. Use the product rule.

b. Rewrite the original function as a sum of powers and differentiate again.

c. Show that the answers are equivalent.

Exercise 7 Solution

Differentiate

$$f(x) = \sqrt{x}(x^2 + 4), \qquad x > 0.$$

a. Use the product rule.

b. Rewrite the original function as a sum of powers and differentiate again.

c. Show that the answers are equivalent.

By the product rule,

$$f'(x) = \frac{x^2 + 4}{2\sqrt{x}} + 2x\sqrt{x}$$

$$= \boxed{\frac{5x^2 + 4}{2\sqrt{x}}}.$$

Expanding into powers first,

$$f(x) = x^{5/2} + 4x^{1/2},$$

so

$$f'(x) = \tfrac{5}{2}x^{3/2} + 2x^{-1/2}.$$

Writing this over the denominator $2\sqrt{x}$ gives the same answer.

Correcting common errors Exercise 8

A student writes:

a. $\frac{d}{dx}(x^2x^3) = (2x)(3x^2) = 6x^3.$

b. $\frac{d}{dx}\left(\frac{x^2+1}{x+1}\right) = \frac{2x}{1}.$

c. $\frac{d}{dx}(7x^3 + 5) = 21x^2 + 5.$

For each statement:

- Identify the error.
- Give the correct derivative.
- State the relevant differentiation rule.

Correcting common errors Exercise 8 Solution

A student writes:

a. $\frac{d}{dx}(x^2x^3) = (2x)(3x^2) = 6x^3.$

b. $\frac{d}{dx}\left(\frac{x^2+1}{x+1}\right) = \frac{2x}{1}.$

c. $\frac{d}{dx}(7x^3 + 5) = 21x^2 + 5.$

For each statement:

- Identify the error.
- Give the correct derivative.
- State the relevant differentiation rule.

a. The derivative of a product is not the product of the derivatives:

$$\frac{d}{dx}(x^2x^3) = (2x)x^3 + x^2(3x^2) = \boxed{5x^4}.$$

b. The derivative of a quotient is not the quotient of the derivatives:

$$\frac{d}{dx}\left(\frac{x^2+1}{x+1}\right) = \boxed{\frac{x^2 + 2x - 1}{(x+1)^2}}, \qquad x \neq -1.$$

c. The derivative of a constant is zero:

$$\frac{d}{dx}(7x^3 + 5) = \boxed{21x^2}.$$

Exercise 9

A moving cart has position

$$s(t) = t^4 - 4t^3 + 6t^2,$$

in metres, for $0 \le t \le 3$ seconds.

a. Find velocity $v(t) = s'(t)$.

b. Find acceleration $a(t) = s''(t)$.

c. Find jerk $j(t) = s'''(t)$.

d. Evaluate all three at $t = 1$, including units.

Exercise 9 Solution

A moving cart has position

$$s(t) = t^4 - 4t^3 + 6t^2,$$

in metres, for $0 \le t \le 3$ seconds.

a. Find velocity $v(t) = s'(t)$.
b. Find acceleration $a(t) = s''(t)$.
c. Find jerk $j(t) = s'''(t)$.
d. Evaluate all three at $t = 1$, including units.

$$\boxed{\begin{aligned} v(t) &= 4t^3 - 12t^2 + 12t, \\ a(t) &= 12t^2 - 24t + 12, \\ j(t) &= 24t - 24. \end{aligned}}$$

At $t = 1$,

$$\boxed{v(1) = 4\ \text{m/s}, \qquad a(1) = 0\ \text{m/s}^2, \qquad j(1) = 0\ \text{m/s}^3.}$$

Exercise 10

A small vehicle's position is

$$s(t) = t^3 - 6t^2 + 9t, \qquad 0 \le t \le 4,$$

where s is in metres and t in seconds.

a. Find its average velocity over $[0, 2]$.

b. Find its instantaneous velocity at $t = 2$.

c. Find all times when it is at rest.

Exercise 10 Solution

A small vehicle's position is

$$s(t) = t^3 - 6t^2 + 9t, \qquad 0 \le t \le 4,$$

where s is in metres and t in seconds.

a. Find its average velocity over $[0, 2]$.
b. Find its instantaneous velocity at $t = 2$.
c. Find all times when it is at rest.

a. Since $s(0) = 0$ and $s(2) = 2$,

$$v_{\text{avg}} = \frac{2-0}{2-0} = \boxed{1 \text{ m/s}}.$$

b. $v(t) = 3t^2 - 12t + 9 = 3(t-1)(t-3)$,

so

$$\boxed{v(2) = -3 \text{ m/s}}.$$

c. The vehicle is at rest when

$$\boxed{t = 1 \text{ s and } t = 3 \text{ s}}.$$